FOREST NETRA

Wiring and Communication Guide

Sensor nodes • Bluetooth Classic • LoRa gateway • Cloud telemetry

1. System at a glance

Forest Netra uses three ESP32 sensor nodes. Each sensor node collects its own measurements and exposes a compact packet through Bluetooth Classic Serial Port Profile (SPP). The ESP32 + SX1262 TX module connects to the nodes one at a time, combines the available readings, and transmits one compact LoRa packet to the ESP32 + SX1262 RX gateway. The RX gateway forwards telemetry over Wi-Fi/Internet to Adafruit IO for temporary cloud monitoring.

Step	Unit	Connection / purpose
1	Soil node	JXBS 7-in-1 → RS485-to-TTL → ESP32
2	Tree node	TCS34725 → I^2C → ESP32
3	Environmental node	BMP390 + DHT11 + BH1750 → ESP32
4	TX / aggregator	Bluetooth Classic SPP from each node → combine payload → LoRa
5	RX / gateway	LoRa receive → Wi-Fi/Internet → Adafruit IO

Communication direction: Soil / Tree / Environmental nodes → Bluetooth Classic → TX → LoRa 865 MHz → RX → Wi-Fi/Internet → Adafruit IO.

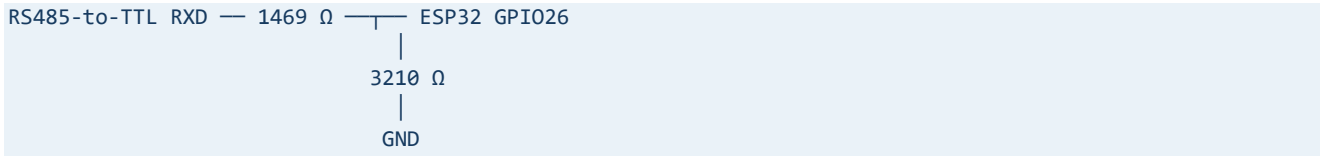
2. Soil node wiring

Main hardware: ESP32, JXBS 7-in-1 soil sensor, RS485-to-TTL interface, 12 V battery, and a DC-DC buck converter.

JXBS sensor wire	Connect to
Brown	+12 V from the 12 V battery
Black	GND of the 12 V supply
Yellow	RS485 A+ on the interface
Blue	RS485 B− on the interface

Power / interface connection	Recorded connection
12 V battery	Powers the JXBS soil sensor and feeds the buck converter input.
Buck converter output	Approximately 4.83 V in the documented setup.
Buck + / −	RS485-to-TTL VCC / GND, respectively.
RS485-to-TTL TXD	ESP32 GPIO27.
RS485-to-TTL RXD	Through the resistor divider to ESP32 GPIO26.
UART setting	9600 baud; automatic-direction RS485-to-TTL interface.

RX voltage divider



The supplied reference estimates the divider output as 4.83 × 3210 / (1469 + 3210) ≈ 3.31 V.

Soil packet over Bluetooth

- Bluetooth device name: ForestNetra_Soil
- Bluetooth mode: Classic Bluetooth SPP server
- The packet fields represent moisture, soil temperature, electrical conductivity, pH, nitrogen, phosphorus, and potassium.

```
M=...,ST=...,E=...,PH=...,N=...,PP=...,K=...
```

Power note: The source PDF does not specify the exact power connection for the Soil ESP32. Do not infer an unrecorded power path.

3. Tree node wiring

Main hardware: ESP32 and TCS34725 RGB/clear optical sensor. The sensor provides raw colour information used by the project to calculate a greenness proxy (GLI).

TCS34725 pin	Recorded connection
SDA	ESP32 D1
SCL	ESP32 D2
VCC	ESP32 3V3
GND	ESP32 GND
INT	Not connected

- Optical sensor settings in the reference: 50 ms integration time and 4X gain.
- GLI is an optical greenness indicator, not a direct chlorophyll measurement.
- Bluetooth device name: ForestNetra_Tree; mode: Classic Bluetooth SPP server.

```
TREE|C=<raw_c>,G=<gli,4>
```

4. Environmental node wiring

Main hardware: ESP32, BMP390 pressure/temperature sensor, DHT11 humidity sensor, and BH1750 ambient-light sensor.

Sensor pin	Recorded connection
BMP390 VCC / GND	3.3 V / GND
BMP390 SDA / SCL	D1 / GPIO5 and D2 / GPIO4
BMP390 INT	Not connected
BH1750 VCC / GND	3.3 V / GND
BH1750 SDA / SCL	D1 / GPIO5 and D2 / GPIO4; shared I^2C bus with BMP390
BH1750 ADDR	GND
DHT11 VCC / GND	3.3 V / GND
DHT11 DATA	D5 / GPIO14

- BMP390 provides temperature and pressure; altitude is calculated from pressure.
- DHT11 provides relative humidity (and temperature); the current telemetry packet uses humidity.
- BH1750 measures ambient light in lux.
- Bluetooth device name: ForestNetra_Environmental; mode: Classic Bluetooth SPP server.

```
ENV|T=<BMP temp>,P=<pressure hPa>,A=<altitude>,H=<humidity>,L=<lux>
```

Pin-label note: The wiring names above reproduce the supplied PDF, including D1/D2 labels. Confirm how those labels map to GPIO on the exact ESP32 board being used before rewiring.

5. TX node: Bluetooth aggregation and LoRa wiring

Main hardware: ESP32 + SX1262 LoRa module. The TX node acts as Bluetooth master/client and LoRa transmitter.

How TX collects node data

- Connect to ForestNetra_Soil using its Bluetooth MAC address, receive the soil packet, and disconnect.
- Connect to ForestNetra_Tree, receive the tree packet, and disconnect.
- Connect to ForestNetra_Environmental, receive the environmental packet, and disconnect.
- Combine the available fields into one compact telemetry message.
- Transmit the combined message as one LoRa packet.

Sensor node	Bluetooth name	MAC address
Soil	ForestNetra_Soil	84:0D:8E:0F:4E:2A
Tree	ForestNetra_Tree	A0:A3:B3:29:C3:BE
Environmental	ForestNetra_Environmental	A4:CF:12:9E:45:0A

SX1262 to ESP32 wiring

SX1262 signal	ESP32 GPIO
NSS / CS	GPIO5
SCK	GPIO18
MOSI	GPIO23
MISO	GPIO19
DIO1	GPIO26
RESET	GPIO14
BUSY	GPIO27

LoRa radio settings

Parameter	Value
Frequency	865.0 MHz
Bandwidth	125.0 kHz
Spreading factor	SF7
Coding rate	4/5
Sync word	0x34
TX power	14 dBm
Preamble length	8

Combined telemetry format

```
T=25.03,P=1003.25,A=83.58,H=77.00,L=467.50,C=1306,G=0.1021,M=0.30,ST=27.50,E=0,PH=7.00,N=0,PP=0,K=0
```

The TX collects these source fields: T/P/A/H/L from the environmental node; C/G from the tree node; and M/ST/E/PH/N/PP/K from the soil node.

Missing node behaviour: If a node fails to connect or provide data in a collection cycle, its fields remain zero and TX still transmits the combined packet. A zero can therefore mean “no valid value received,” not necessarily a measured physical zero.

6. RX gateway: LoRa to cloud

Main hardware: ESP32 + SX1262. The RX continuously listens for LoRa packets and receives the full compact telemetry message sent by TX.

Stage	Interface	Action

Stage	Interface	Action
TX → RX	LoRa at 865 MHz	One complete combined telemetry message is sent and received as a single packet.
RX → network	Wi-Fi / Internet	The receiver provides the gateway connection to the internet.
Network → cloud	Internet / Adafruit IO	The received telemetry is sent to Adafruit IO for temporary monitoring.

- The source reference shows the RX reporting packet length, RSSI, and SNR in serial output.
- Example RX diagnostics: message length 99 bytes, RSSI −47 dBm, SNR 12.5 dB. These are example readings, not fixed values.

```
TX → LoRa 865 MHz → RX → Wi-Fi / Internet → Adafruit IO
```

7. Quick connection checklist

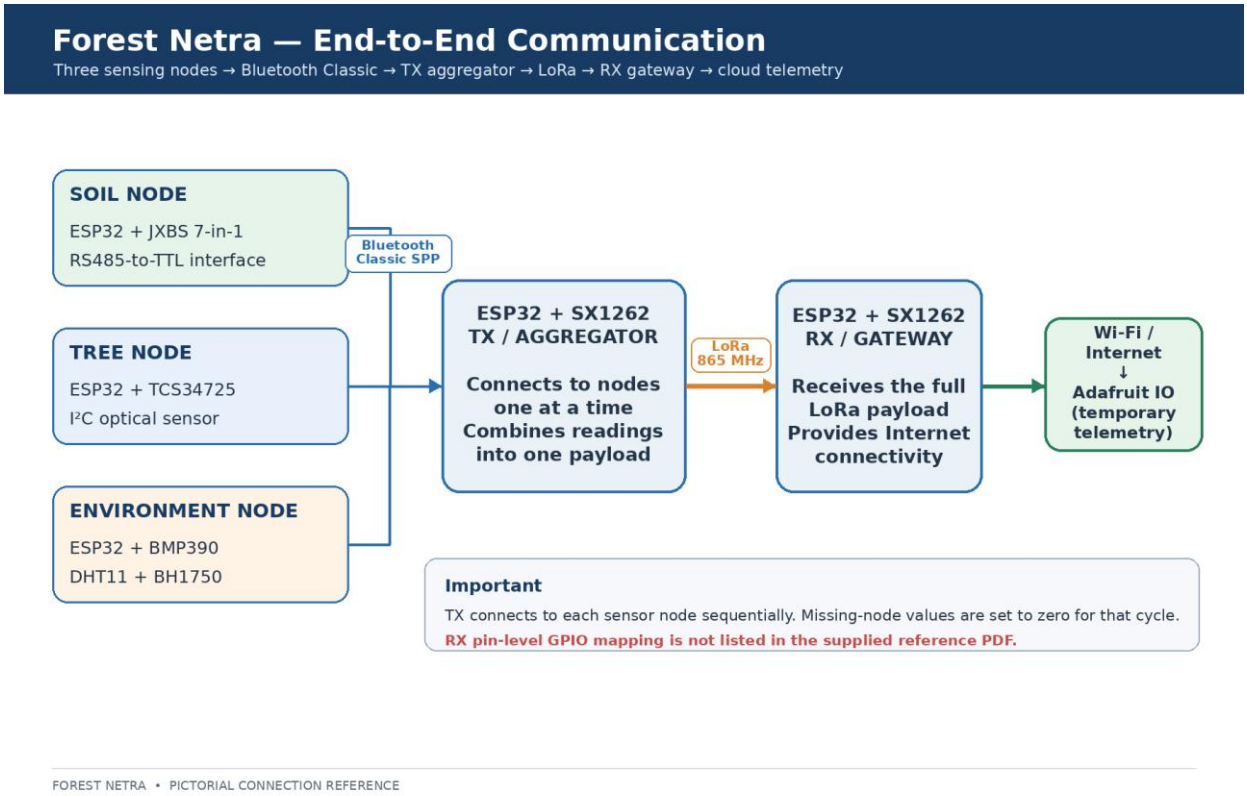
- Soil node: verify 12 V polarity, RS485 A/B, buck output, common ground, and GPIO27/GPIO26 UART wiring including the resistor divider.
- Tree node: verify TCS34725 SDA, SCL, 3V3, GND, and leave INT disconnected as documented.
- Environmental node: verify BMP390 and BH1750 share the recorded I^2C lines; connect BH1750 ADDR to GND; connect DHT11 DATA to GPIO14.
- TX/RX: use matching LoRa radio settings on both modules. Confirm exact board pin mapping against the hardware and source code before wiring.
- Test each sensor node packet over Bluetooth before testing sequential collection at TX.
- Verify the received packet at RX before checking Adafruit IO.

Source basis: Forest Netra — System Hardware & Communication Reference, supplied PDF (9 pages). This guide condenses the wiring and communication details from that document; it does not independently validate the physical connections.

6. Pictorial connection diagrams

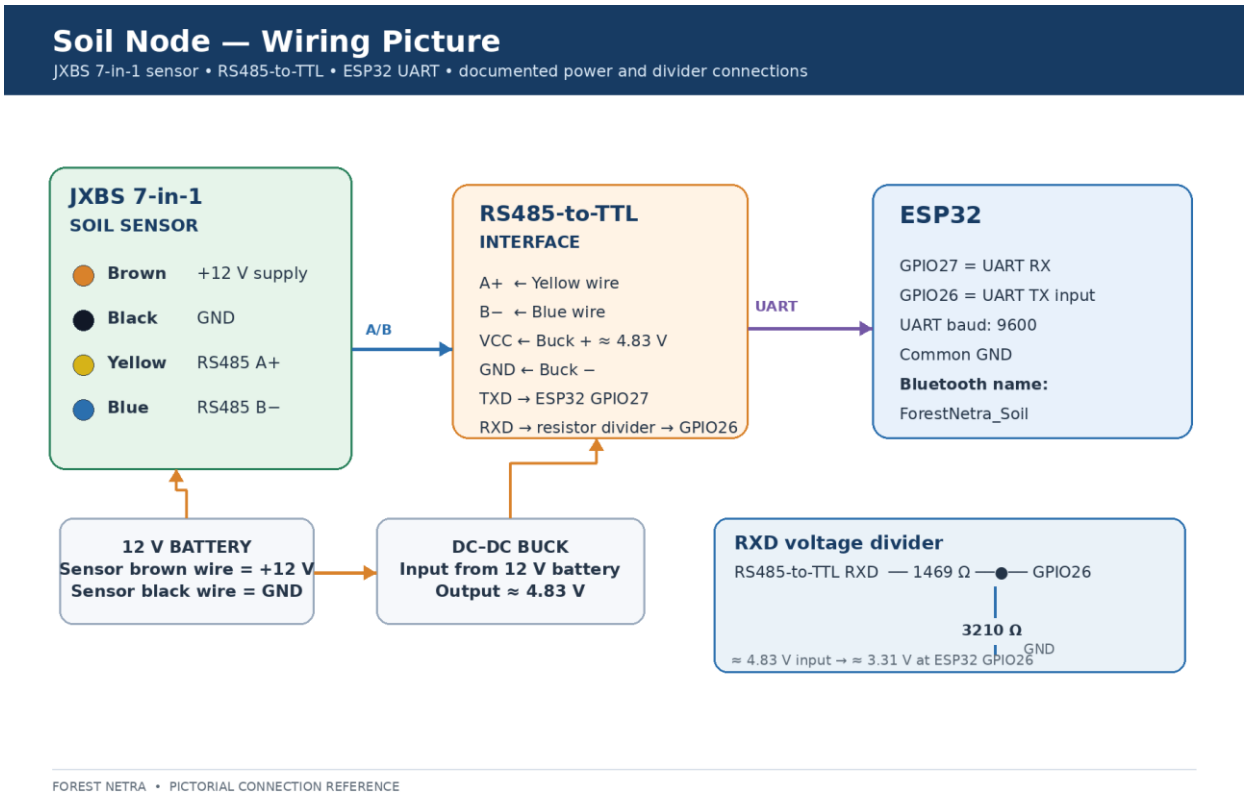
The diagrams below visualize the documented wiring and communication paths. They supplement the connection tables above; use the tables and source PDF as the written pin-level reference.

Figure 1. End-to-end communication architecture



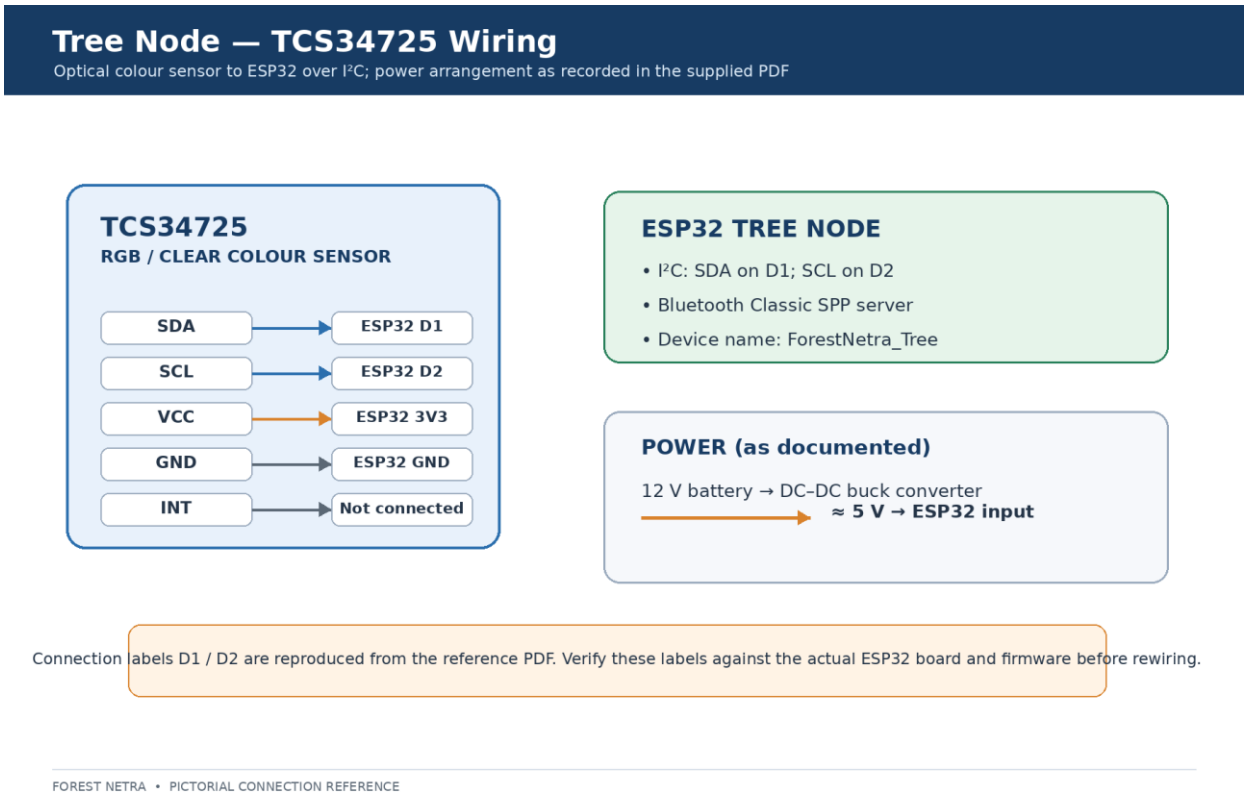
Shows how the three sensing nodes send packets to the TX aggregator over Bluetooth Classic, how the TX forwards a combined payload over LoRa, and how the RX gateway forwards telemetry to Adafruit IO.

Figure 2. Soil node wiring



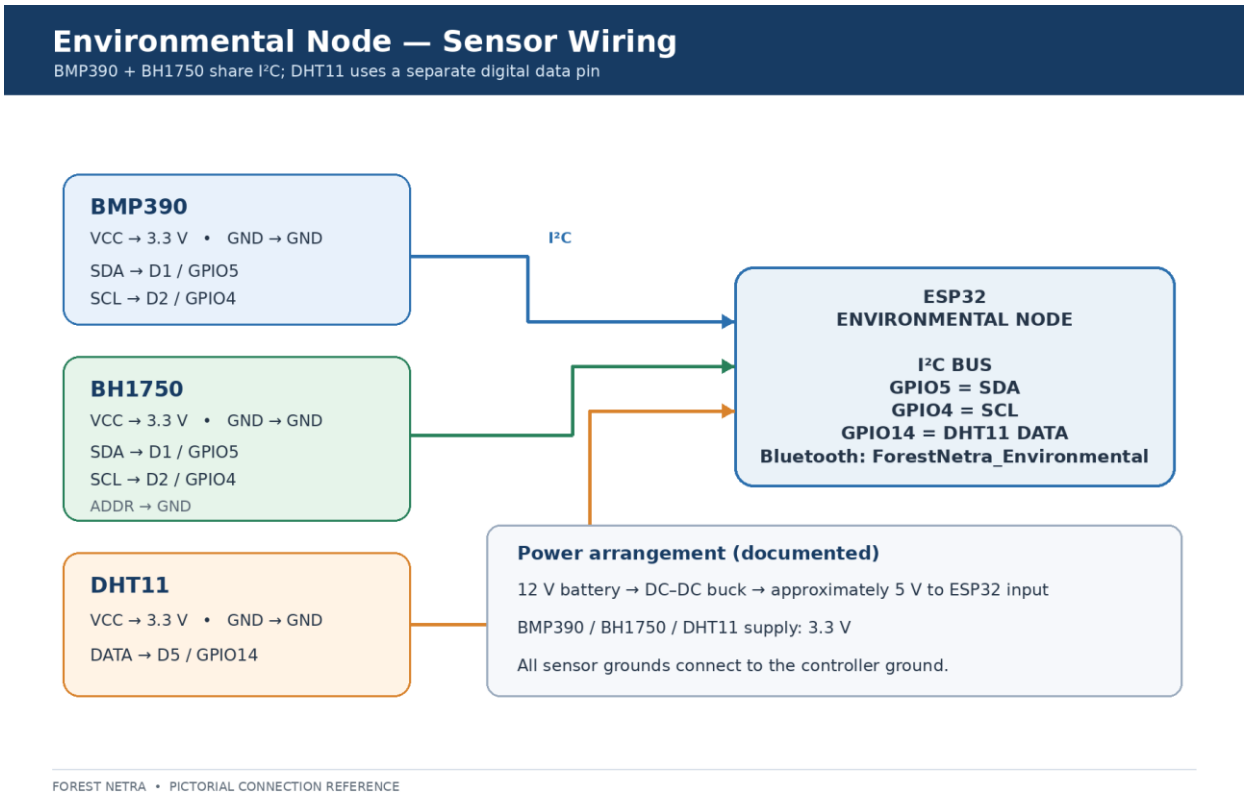
Shows the JXBS sensor wires, RS485 A/B wiring, UART lines, buck-converter supply, and GPIO26 voltage divider. The ESP32 power path itself is not specified by the source PDF.

Figure 3. Tree node wiring



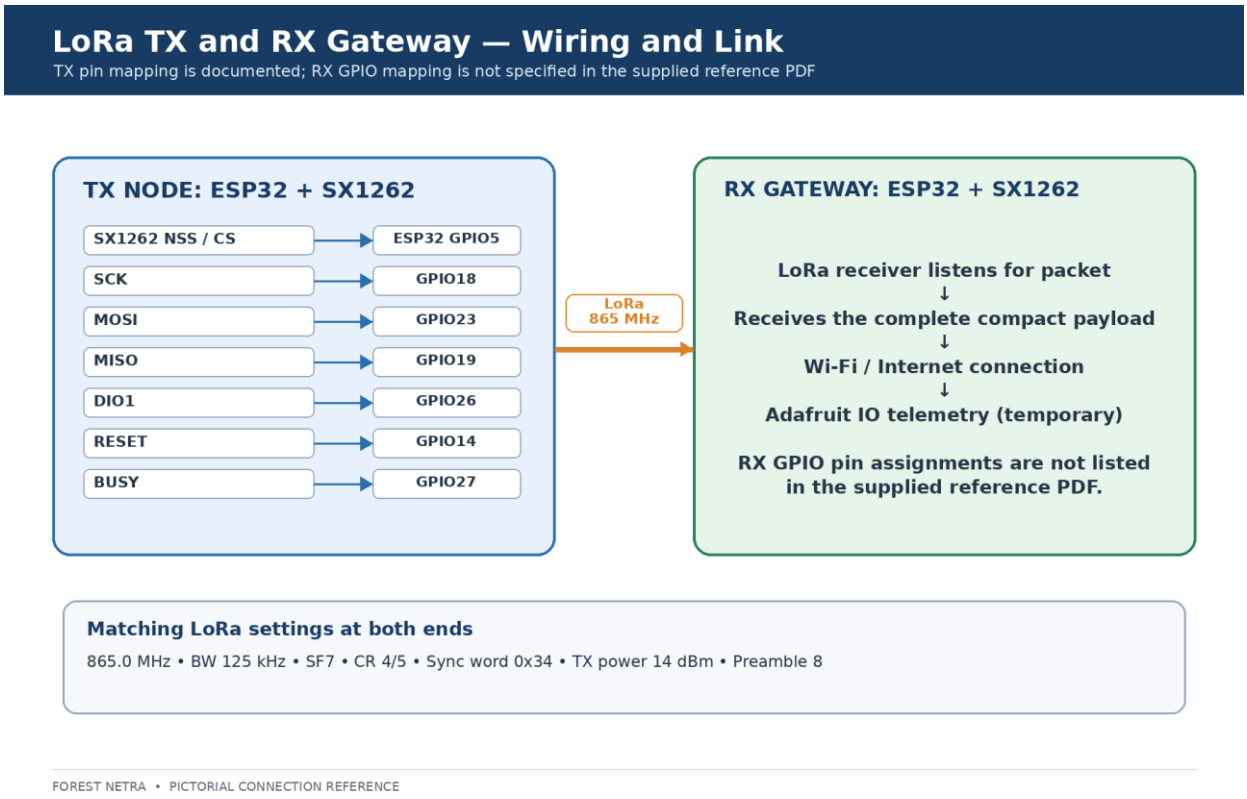
Shows TCS34725 SDA, SCL, VCC, GND, and unused INT connections, plus the documented ESP32 power arrangement.

Figure 4. Environmental node wiring



Shows the shared I²C bus for BMP390 and BH1750, the BH1750 address pin, DHT11 data pin, and the documented power arrangement.

Figure 5. LoRa TX wiring and RX gateway flow



Shows the documented SX1262-to-TX ESP32 GPIO mapping and the TX-to-RX LoRa link. RX GPIO pin assignments are intentionally not inferred because they are absent from the source PDF.